CSC 402 — Mobile Application Programming

Lab 1: Exercises

Task 1 — Create the Project

Project Information

- **Application Name:** CSC402 Lab1
- **Package Name:** 2240006228.iau.edu.sa
- **Language:** Kotlin

Task 2 — Project Tour

Project Tour Table

File / Folder	Purpose
`AndroidManifest.xml`	Declares the application components, permissions, and the launcher activity.
`MainActivity.kt`	Contains the main Activity and the Jetpack Compose user interface.
`res/drawable/`	Contains drawable and image resources used by the application.
`res/mipmap/`	Contains the application launcher icons.
`res/values/strings.xml`	Contains user-visible text and string resources.
`ui/theme/Color.kt`	Defines colors used by the application theme.
`ui/theme/Theme.kt`	Defines the application's theme.

File / Folder	Purpose
`ui/theme/Type.kt`	Defines typography styles.
`build.gradle.kts (Project)`	Configures project-level build plugins and settings.
`build.gradle.kts (Module :app)`	Configures the app module, SDK versions, application ID, version information, and dependencies.
`settings.gradle.kts`	Defines the project structure and included modules.
`gradle/libs.versions.toml`	Stores library and plugin version information.

Q1

Android Studio uses the Android project view to organize files according to their purpose, which makes the project easier to navigate. It groups related files such as Kotlin source files and resources even though they may be stored in different folders on disk.

Task 3 — Read the Manifest

INTERNET Permission Experiment

The `android.permission.INTERNET` permission was temporarily added to the `AndroidManifest.xml`. A screenshot was taken to document the experiment, and the permission was then removed because the application does not use network access.

`android:exported` Experiment

The `android:exported` value of the launcher activity was temporarily changed from `true` to `false`. The application still launched successfully during the experiment, so the value was changed back to `true`.

Q2

The app still launched successfully when `android:exported` was set to `false`. The `exported` attribute controls whether an app component can be launched by other applications or external components.

Task 4 — Read the Gradle Build Files

Gradle Settings Table

Setting	Your Value	What does it control?
`namespace`	`com.example.csc402lab1`	Defines the namespace used for generated and referenced application code.
`applicationId`	`com.example.csc402lab1`	Uniquely identifies the application when it is built and installed.
`compileSdk`	`37`	Specifies the Android API level used to compile the application.
`minSdk`	`24`	Specifies the oldest Android API level that can install the application.
`targetSdk`	`37`	Specifies the Android API level the application is designed and tested for.
`versionCode`	`1`	Identifies the internal version of the application.
`versionName`	`1.1`	Defines the user-visible version name of the application.

Compose / Material3 Version

The project uses **Compose BOM**. The `androidx.compose.material3` dependency does not specify a separate version because its version is managed by the Compose BOM.

Q3

My project has `minSdk = 24`, which allows the application to run on devices using Android 7.0 (API 24) or newer. If I raise `minSdk` to 34, devices running Android versions below API 34

would no longer be able to install the application, but I would gain access to newer Android APIs and would have less need to support older Android versions.

Task 5 — Build and Run the App

First Run

The application was run on the Pixel 6 emulator and displayed: **Hello Android!**

Second Run

The greeting was changed and the application was run again. It displayed: **Hello fatimah!**

Q4

The second build took **781 ms**. The second build was significantly faster because Gradle reused previously compiled and unchanged files instead of rebuilding the entire project from scratch.

Task 6 — String Resources

The user-visible text was moved from the Kotlin code into `res/values/strings.xml`.

The following resources were added:

- `app_name` = `CSC 402 Lab 1`
- `welcome_message` = `Welcome to Mobile Application Programming`
- `student_name` = `Fatimah Al Dhahi`

The final application displays:

- **Welcome to Mobile Application Programming**
- **Fatimah Al Dhahi**

AI Use & Disclosure Log

Tool / Model	Task	What I Asked	What I Changed
Gemini / AI	Lab 1	Asked for guidance on structuring the lab report and explaining build outputs.	Organized tables, questions, and answers according to course guidelines.